# AI Butler-Garbage Sorting

Before running the function, you need to close the App and large programs. For the closing method, refer to [4. Preparation] - [1. Manage APP control services].

## 1. Function Description

After the program starts, give the large model a garbage sorting instruction. After thinking, the large model will control the robotic arm to sort garbage label code blocks on the table according to requirements.

## 2. Startup

After entering the Docker container, enter the following commands in the Docker terminal.

```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

Then open a second terminal and input the following command,

```
ros2 run text_chat text_chat
```

Garbage sorting has the following instructions:

- Sort a single type of garbage: for example, sort toxic garbage
- Keep a certain type of garbage: for example, sort all garbage except recyclable garbage
- Sort all garbage on the desktop: for example, clean up all garbage on the desktop

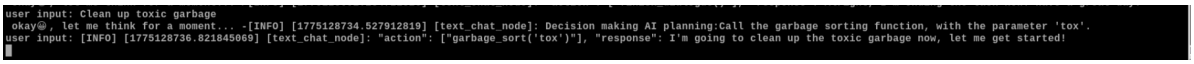
### 2.1. Sort a Single Type of Garbage

#### 2.1.1. Startup

Input in the terminal that started text_chat,

```
Clean up toxic garbage
```

Press Enter. If it's a voice version, wake up the voice module and directly say "Clean up toxic garbage" to the voice module. Wait for the large model to think and reply, as shown in the figure below,




The robotic arm will pick up the toxic garbage label code blocks on the desktop.

#### 2.1.2. Task Planning

Call the garbage sorting function, with the parameter 'tox'.

#### 2.1.3. Core Source Code Analysis

The garbage_sort function is located at `LargeModel_ws/src/largemodel/largemodel/action_service.py`. The function takes a type_ argument, which is used to determine the type of garbage to sort.

```
def garbage_sort(self, type_):
    Arm.Arm_serial_servo_write6(90,120,10,10,90,30,1000)
    time.sleep(1.0)
    type_ = type_.strip("'\"")  # Remove single and double quotes
    if type_ == "rec":
        target_type_ = float(1)
    elif type_ == "tox":
        target_type_ = float(2)
    elif type_ == "wet":
        target_type_ = float(3)
    elif type_ == "dry":
        target_type_ = float(4)
    else:
        self.get_logger().info(
            "Fatal ERROR:Incorrect type_ input,Does the AI output not meet
expectations?"
        )
        self.action_status_pub(
            "garbage_sort_failed", type_=type_
        )
        return
    # Start robotic arm grasping program
    cmd1 = "ros2 run dofbot_pro_yolov11 yolov11_sortation"
    # Start image conversion program
    cmd2 = "ros2 run dofbot_pro_yolov11 msgToimg"
    # Start garbage recognition program
    cmd3 = f"ros2 run largemodel_arm yolov11_ALM --ros-args  -p target_type:=
{target_type_:.1f}"

    subprocess.Popen(
        [
            "gnome-terminal",
            "--title=grasp_desktop_garbage_sort",
            "--",
            "bash",
            "-c",
            f"{cmd1}; exec bash",
        ]
    )

    subprocess.Popen(
        [
            "gnome-terminal",
            "+--title=detect_img",
            "--",
            "bash",
            "-c",
            f"{cmd2}; exec bash",
        ]
    )

    subprocess.Popen(
        [
            "gnome-terminal",
            "--title=garbage_detect",
            "--",
            "bash",
            "-c",
```

```
            f"{cmd3}; exec bash",
        ]
    )
```

Main analysis of the garbage recognition program yolov11_ALM, source code path: `LargeModel_ws/src/largemodel_arm/largemodel_arm/yolov11_ALM.py`

```
# Get the garbage type to be sorted through command line arguments
self.declare_parameter('target_type', 1.0)
self.TargetType =
int(self.get_parameter('target_type').get_parameter_value().double_value)
# Modify self.waste according to the value of self.TargetType, which contains all
garbage names of this type
if self.TargetType == 1:
    self.waste = ['Newspaper','Zip_top_can','Book','Old_school_bag']
elif self.TargetType == 2:
    self.waste =
['Syringe','Expired_cosmetics','Used_batteries','Expired_tablets']
elif self.TargetType == 3:
    self.waste = ['Fish_bone','Egg_shell','Apple_core','Watermelon_rind']
elif self.TargetType == 4:
    self.waste =
['Toilet_paper','Peach_pit','Cigarette_butts','Disposable_chopsticks']

# Initialize a YOLO object detection model instance, load precompiled TensorRT
engine file (best.engine), and specify the model task type as object detection
(detect)
self.yolo_model =
YOLO("/home/jetson/dofbot_pro_ws/src/dofbot_pro_yolov11/dofbot_pro_yolov11/best.
engine", task='detect')
# Create a subscriber to subscribe to the converted image data topic
self.image_sub =
self.create_subscription(ImageMsg,"/image_data",self.image_sub_callback,qos_prof
ile=1)

# In the callback function image_sub_callback
# Use YOLO11 for object detection
results = self.yolo_model(self.img, save=False, verbose=False)
boxes = results[0].boxes
# Traverse detection results
for box in boxes:  # detections per image
    x_min, y_min, x_max, y_max = map(int, box.xyxy[0])
    class_id = int(box.cls)
    confidence = float(box.conf)
    label = f"{self.yolo_model.names[class_id]} {confidence:.2f}"
    # Calculate center of gravity position
    center_x = (x_min + x_max) // 2
    center_y = (y_min + y_max) // 2
    # Extract the information of the currently detected label code and assign it
to center, including center point coordinates and label code name
    center = Yolov11Detect()
    center.centerx = float(center_x)
    center.centery = float(center_y)
    center.result = str(self.yolo_model.names[class_id])
    # If the name of the currently detected label code is in the self.waste list,
it means it is the target sorting object, publish sorting topic flag and label
code position information
```

```
    if center.result in self.waste:
        start_flag = Bool()
        start_flag.data = True
        self.pub_SortFlag.publish(start_flag)
        self.detect_flag = True
        self.cnt = self.cnt + 1
        print("Found the target.")
        self.pubDetect.publish(center)
        self.start_flag = False
```

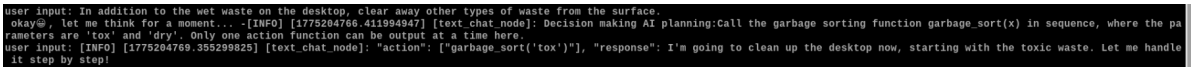
## 2.2. Keep a Certain Type of Garbage

### 2.2.1. Startup

Input in the terminal that started text_chat,

```
In addition to the wet waste on the desktop, clear away other types of waste from
the surface.
```

Press Enter. If it's a voice version, wake up the voice module and directly say "In addition to the wet waste on the desktop, clear away other types of waste from the surface." to the voice module. Wait for the large model to think and reply, as shown in the figure below,




Except for wet garbage label code blocks, the robotic arm will pick up other types of garbage label code blocks on the desktop.

### 2.2.2. Task Planning

Sequentially call the garbage sorting function `garbage_sort(x)`, parameters are 'tox', 'rec' and 'dry'. Here only one action function can be output at a time.

Task steps:

1. Call `garbage_sort('tox')`
2. Call `garbage_sort('rec')`
3. Call `garbage_sort('dry')`

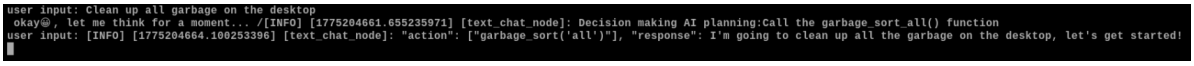
## 2.3. Sort All Garbage on the Desktop

### 2.3.1. Startup

Input in the terminal that started text_chat,

```
Clean up all garbage on the desktop
```

Press Enter. If it's a voice version, wake up the voice module and directly say "Clean up all garbage on the desktop" to the voice module. Wait for the large model to think and reply, as shown in the figure below,




The robotic arm will sort all garbage label blocks on the desktop,

#### 2.3.2. Task Planning

Call the garbage_sort_all() function.